Analysis from Provete et al. 2014. Phylogenetic and taxononomic diversity interactively influence freshwater ecosystem functioning

Diogo B. Provete

April 10, 2014

These are the analyses I run for the Amy's experiment (Downing & Leibold 2002 Nature) in which I investigated the effects of phylogenetic diversity on ecosystem functioning. This is one of my PhD. dissertation chapters.

Loading R packages and data input

```
library(spacodiR)  #0.13.0115
library(ecoPD)  #0.2
library(picante)  #1.6-2
library(apTreeshape)  #1.4-5
library(phytools)  #0.3-93
library(bbmle)  #1.0.16
library(MuMIn)  #1.9.13
library(ggplot2)  #0.9.3.1
# Run in R 3.0.3
```

```
dados <- read.csv("data_diogo.csv", header = T, sep = ",")
head(dados)
```

```
##   tank treat comp   PD    PSV div dist time    dc    ph  prbio  prod
## 1    5     5   11 4901 0.5844   9    6    6 3.513  4.40  21.30 0.144
## 2    1     1    4 3572 0.6729   1    0    5 4.289  1.46  63.41 0.396
## 3   31     4    6 4107 0.6199   1    1    1 7.233  6.72 218.40 0.369
## 4   54     4    7 2496 0.3513   1    1    2 4.652 12.39  60.28 0.352
## 5   65     1    5 3989 0.6199   1    0    3 4.387  1.76  14.70 0.349
## 6   50     1    3 3645 0.7057   1    0    1 6.251  9.60  77.50 0.347
##    prpro  resp
## 1 144.80 0.111
## 2 153.04 0.213
## 3  23.45 0.233
## 4  31.02 0.276
## 5  40.20 0.259
## 6  24.27 0.246
```

```r
attach(dados)

trata = factor(div)
COMP <- factor(comp)
TIME <- factor(time)

me <- aggregate(prod, list(trata, PD), "mean")
dec.plot <- aggregate(dc, list(trata, PD), "mean")
names(dec.plot) = c("Diversity", "PD", "Decomposition")  #renaming the columns to make the
resp.plot <- aggregate(resp, list(trata, PD), "mean")
names(resp.plot) = c("Diversity", "PD", "Decomposition")  #renaming the columns to make th
```

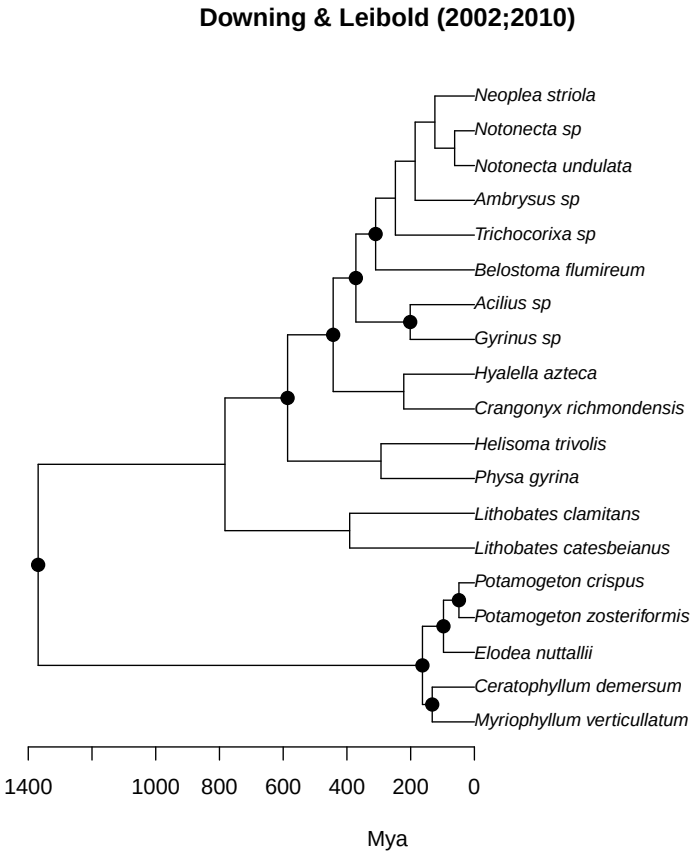
Phylogeny input and calculating PD variables

Here I assembled by hand a phylogeny of all species in the treatments using Mesquite. Then, I looked for node ages in www.timetree.org. After I got these data, I took the tree and the node ages to Phylocom to obtain a pseudochronogram by using BLADJ. From that, I imported the dated tree again into R.

```r
# ---Data input and manipulation of trees to be dated in Phylocom
phylog.2002 = read.tree("undated_tree.txt")
phyloNodes = makeNodeLabel(phylog.2002)
write.tree(phyloNodes, file = "phylo_2002.txt")

# ---Working wih dated phylogenies
amy.nature = read.tree("phylog_dated_2002.txt")
plot(amy.nature, cex = 0.8)
axisPhylo()
nodelabels(node = c(20, 21, 22, 23, 24, 27, 29, 31, 33, 32), bg = "black", frame = "circle
    cex = 0.3)
title("Downing & Leibold (2002;2010)", xlab = "Mya")
```

Figure 1: Dated phylogeny with the 19 species that remained at the end of the experiment. The black dots indicate the nodes with known ages used to date the tree



Tree shape metrics

Since previous studies suggesteed that tree imbalance and rootness could alter the influence of phylogeny on ecosystem functioning, I calculated two commonly used metrics, I_c and $Pybus'\gamma$.

```
# ---Playing with tree shape metrics
treeNature = as.treeshape(amy.nature)
colless.test(treeNature)  #the tree is more balanced than

##
##
##  Test of the yule hypothesis using the Colless index
## Statistic = 49
## Standardized Statistic = 0.7504
```

```
## p-value = 0.86
## alternative hypothesis: the tree is more balanced than predicted by the yule model
##
## Note : the p-value was computed using a Monte-Carlo method

# predicted by the Yule model
z = ltt(amy.nature, plot = F, gamma = F)
set.seed(1001)
gammatest(z)  #gamma=0.40; P=0.686

## $gamma
## [1] 0.4038
##
## $p
## [1] 0.6864
```

Calculating PD variables

Importing data from the supplementary material of Downing & Leibold (2002) that describes final species composition in each treatment to calculate PD.

```
# ---Species composition per treatments, calculating PD
treat = read.table("treat.txt", h = T)

# --Independent on species richness
mpd.nature = mpd(t(treat), cophenetic(amy.nature))  #mpd
obj = phylo4d(amy.nature, tip.data = treat)
cadotte = haed(obj)  #Haed
raodiv = raoD(t(treat), amy.nature)$Dkk

# --Dependent on species richness
PD.nature = picante::pd(t(treat), amy.nature)  #Faith's PD
mntd.nature = mntd(t(treat), cophenetic(amy.nature))  #mnnd
SimpsonPhyl = simpson(obj, "phylogenetic")  #Simpson Phylogenetic
Shannon = diversity(t(treat), "shannon")  #Shannon
Simpson = diversity(t(treat), "simpson")  #Simpson
J = sum(cophenetic.phylo(amy.nature))/(colSums(treat)^2)  #see Schweiger et al. 2008\\ Dec
```

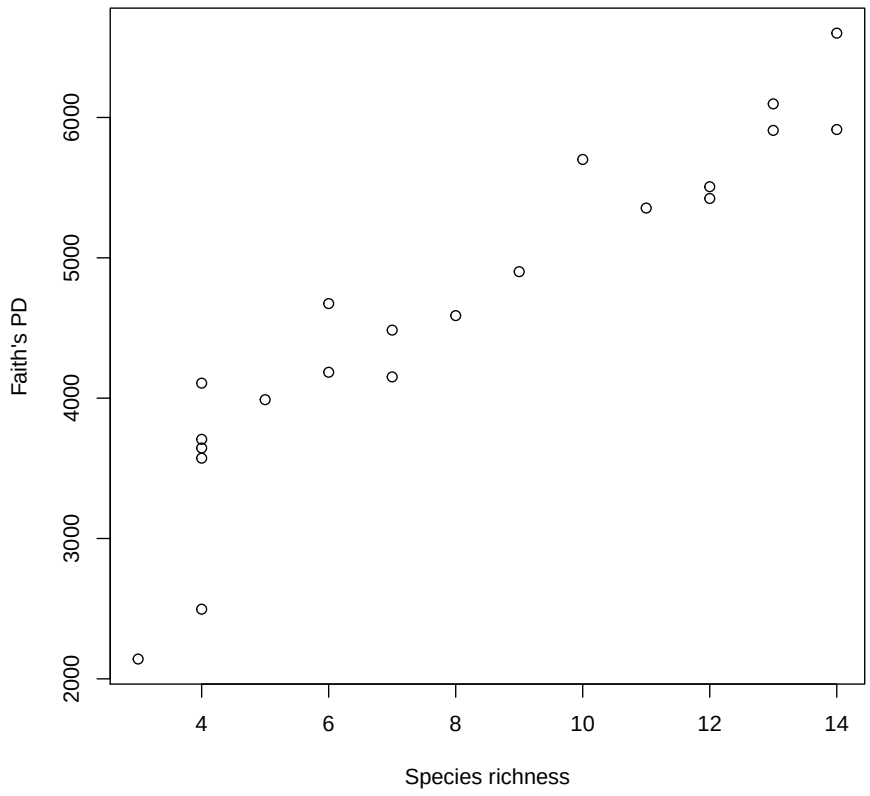
Pasting the new variables into the dataset.

```
# ---Binding phylogenetic diversity variables into data file
d <- dados[order(dados$comp), ]  #sorting data by composition
d$mpd <- rep(mpd.nature, each = 12)  #repeat MPD 12 times
d$haed <- rep(cadotte, each = 12)
d$raodiv <- rep(raodiv, each = 12)
d$mntd <- rep(mntd.nature, each = 12)
d$SimpsonPhyl <- rep(SimpsonPhyl, each = 12)
d$Shannon <- rep(Shannon, each = 12)
d$Simpson <- rep(Simpson, each = 12)
```

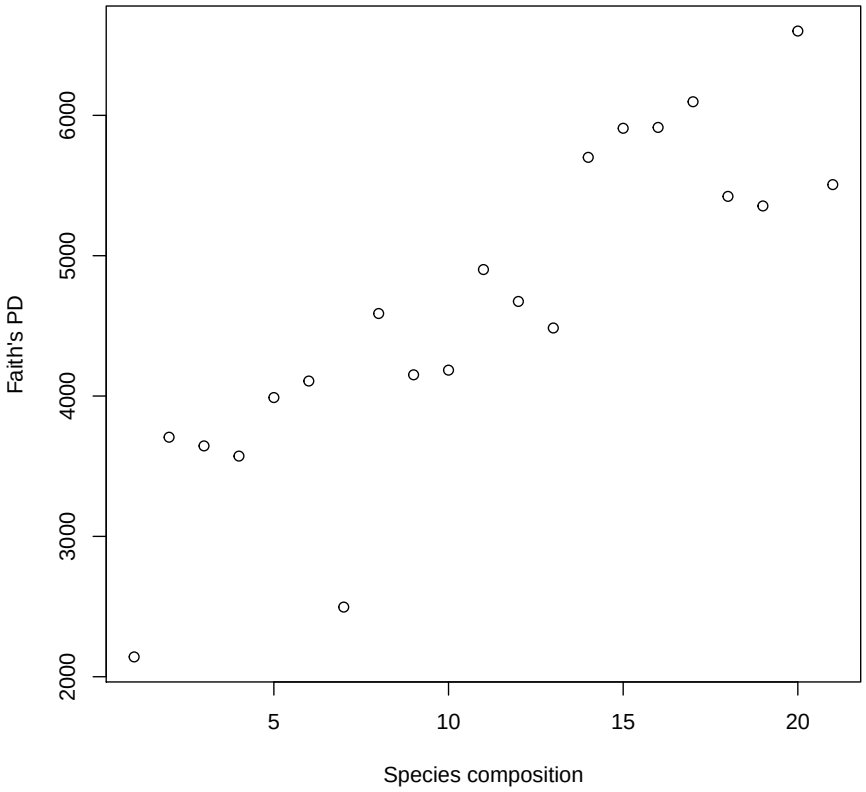
```r
d$Intens_Q <- rep(J, each = 12)
```

Exploring the variation in PD variables in relation to species composition (treatment) and species richness.

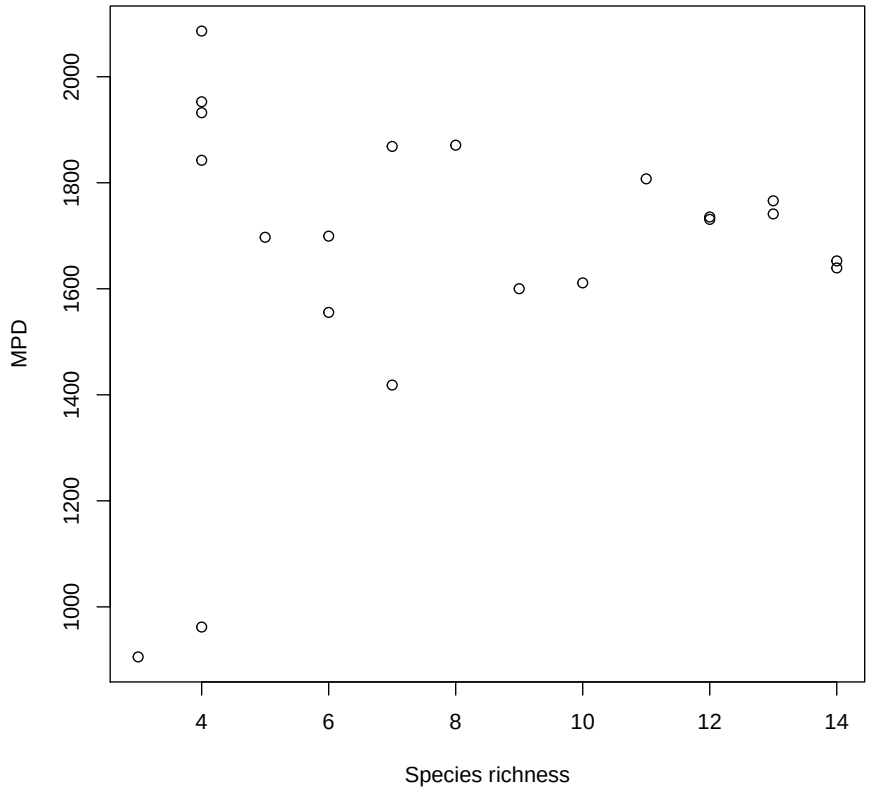
```r
# ---Plots variation in PD and MPD
plot(PD.nature[[1]] ~ PD.nature[[2]], xlab = "Species richness", ylab = "Faith's PD")
```



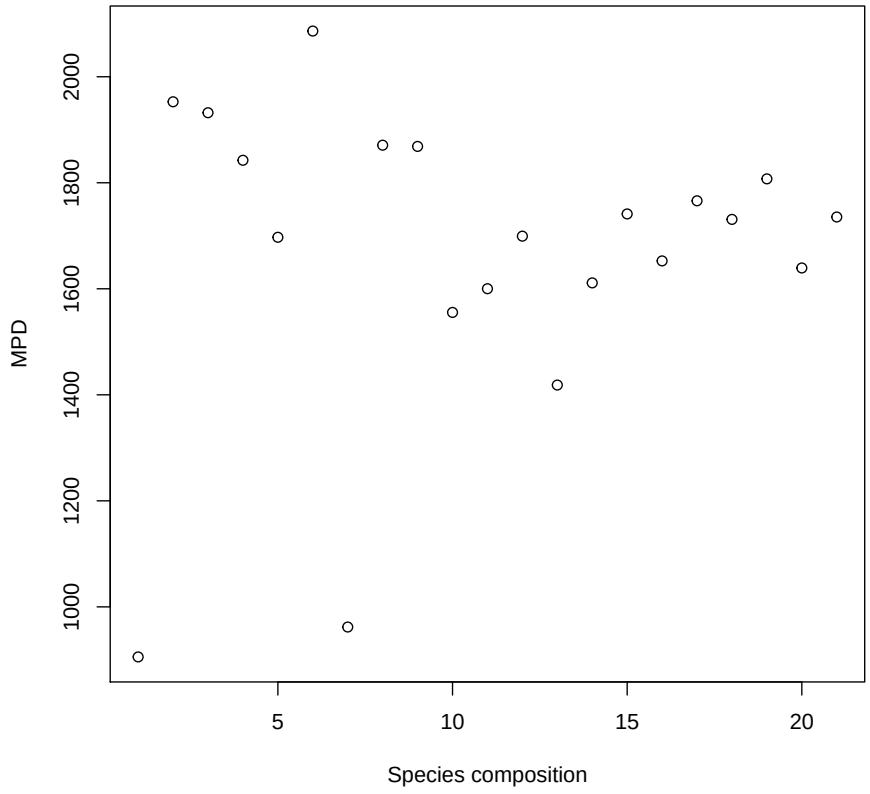

```r
plot(PD.nature[[1]], xlab = "Species composition", ylab = "Faith's PD")
```



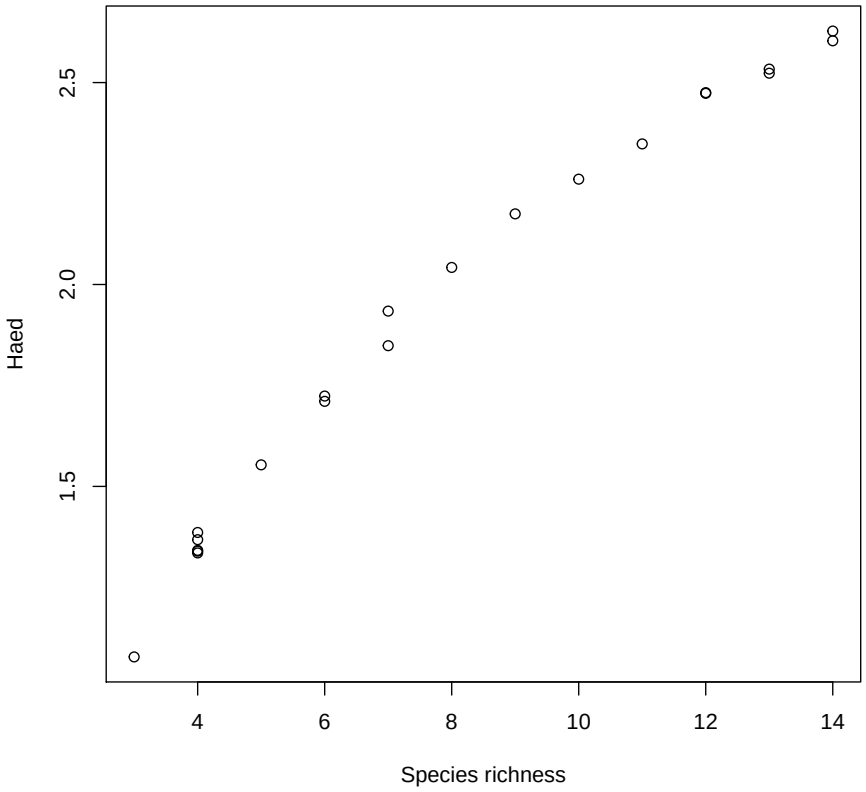

```
plot(mpd.nature ~ PD.nature[[2]], xlab = "Species richness", ylab = "MPD")
```



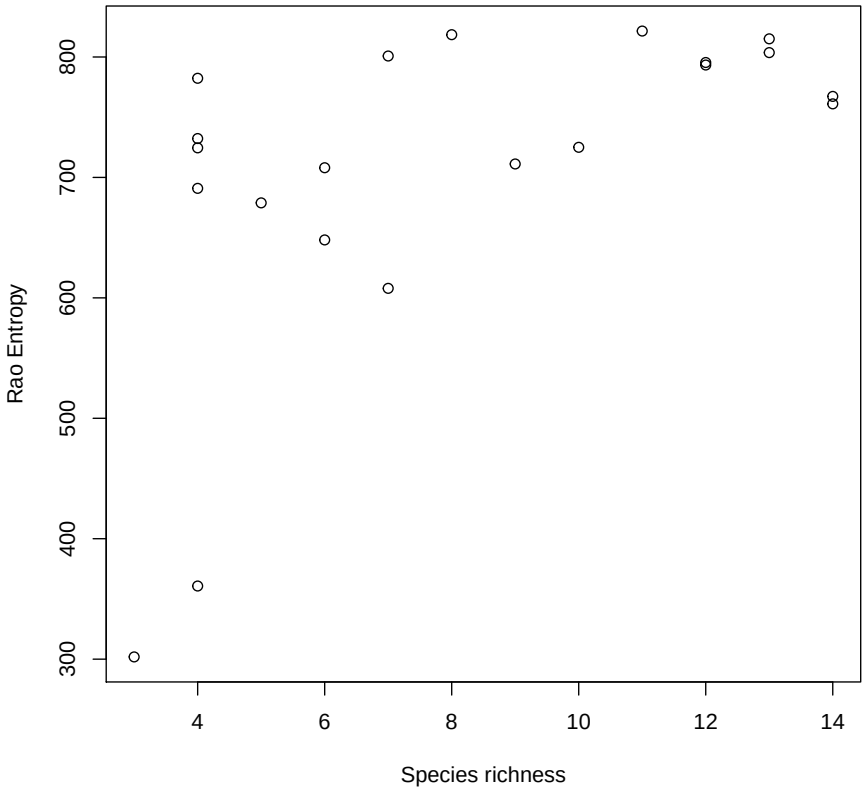

```
plot(mpd.nature, xlab = "Species composition", ylab = "MPD")
```



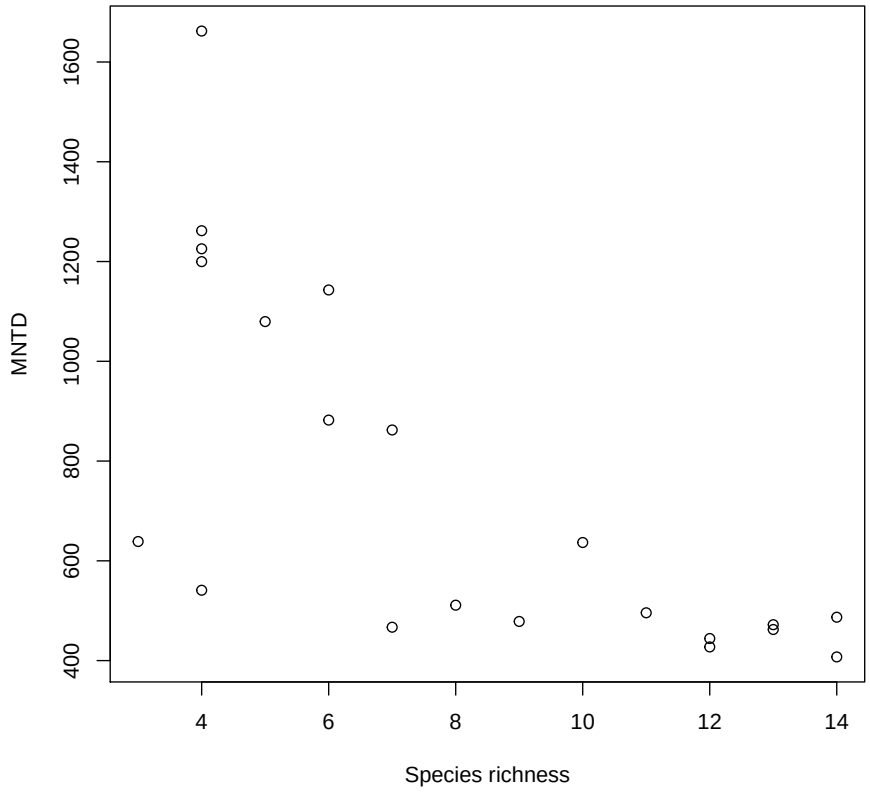

```
plot(cadotte ~ PD.nature[[2]], xlab = "Species richness", ylab = "Haed")
```



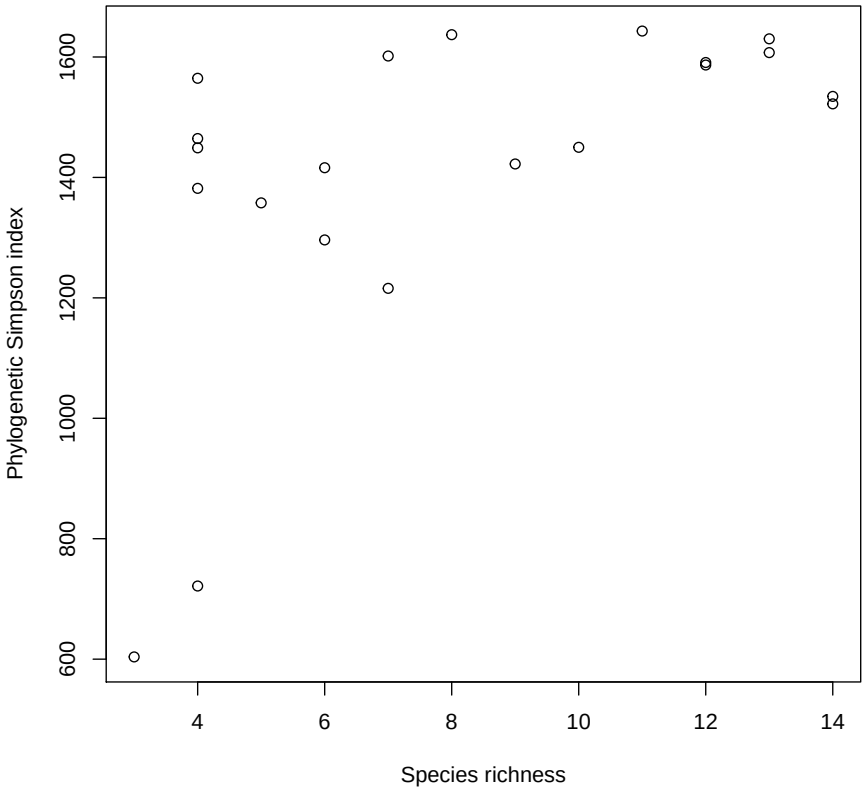

```
plot(raodiv ~ PD.nature[[2]], xlab = "Species richness", ylab = "Rao Entropy")
```



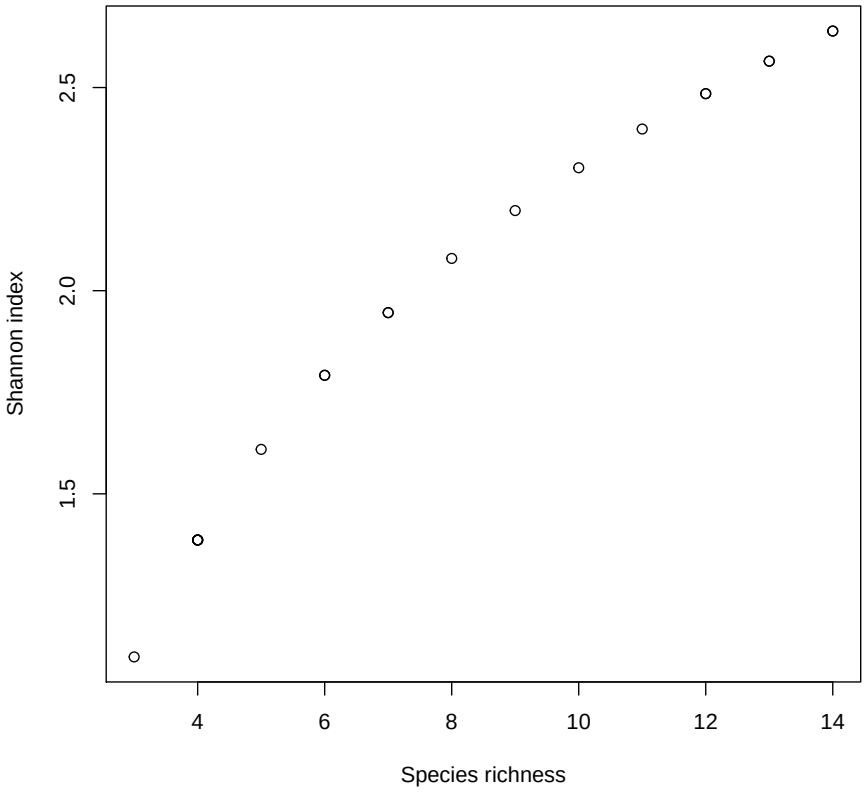

```
plot(mntd.nature ~ PD.nature[[2]], xlab = "Species richness", ylab = "MNTD")
```



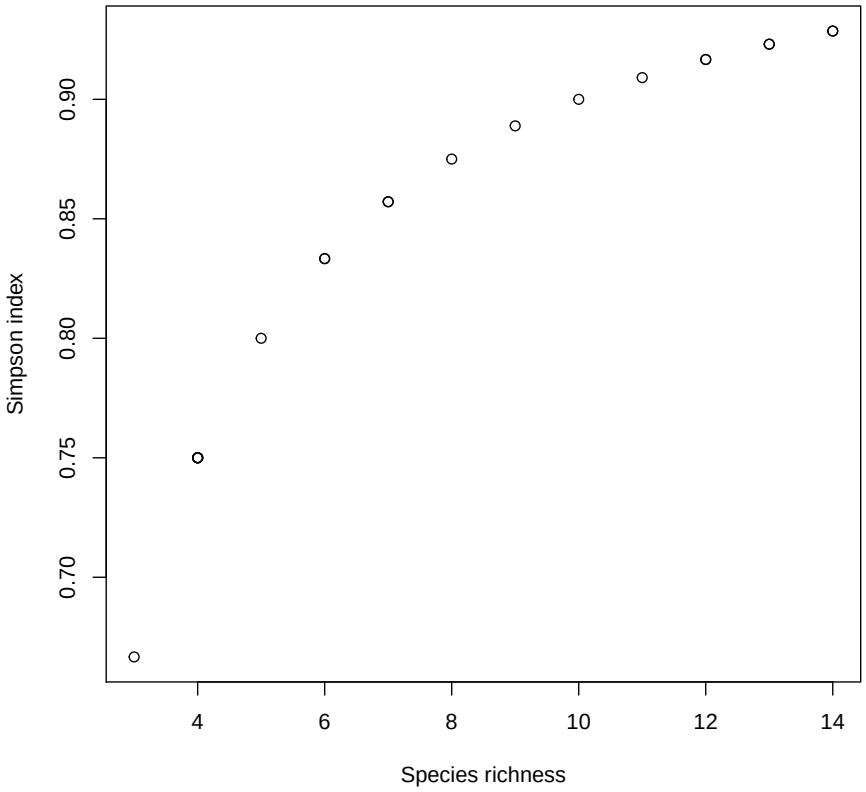

```
plot(SimpsonPhyl ~ PD.nature[[2]], xlab = "Species richness", ylab = "Phylogenetic Simpson
```



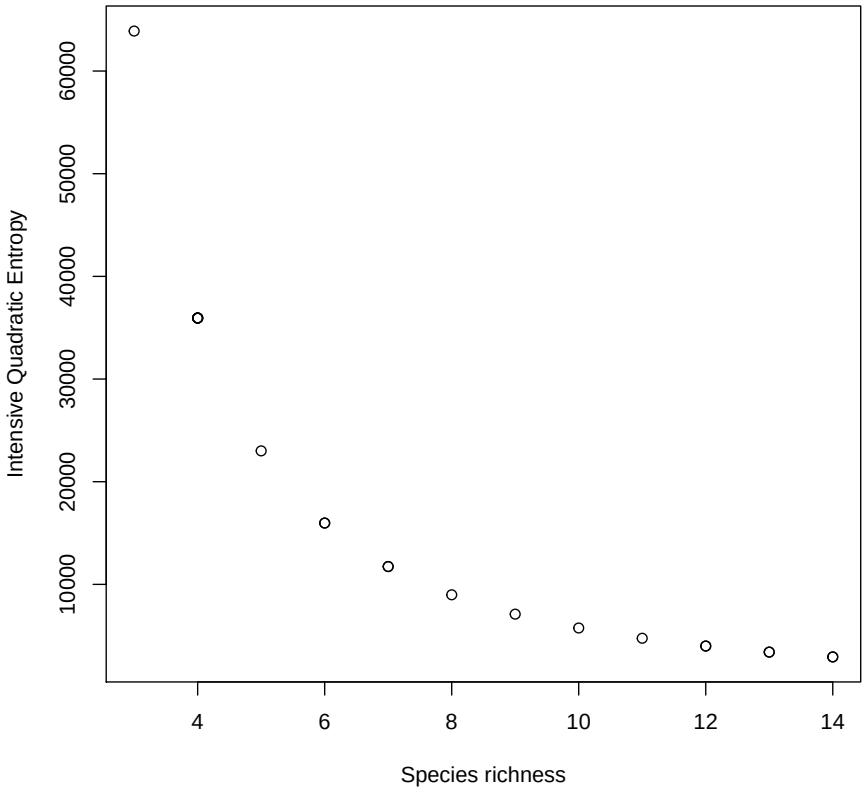

```
plot(Shannon ~ PD.nature[[2]], xlab = "Species richness", ylab = "Shannon index")
```




```
plot(Simpson ~ PD.nature[[2]], xlab = "Species richness", ylab = "Simpson index")
```




```
plot(J ~ PD.nature[[2]], xlab = "Species richness", ylab = "Intensive Quadratic Entropy")
```



Visualizing species incidence in each treatment plotted on the regional phylogeny

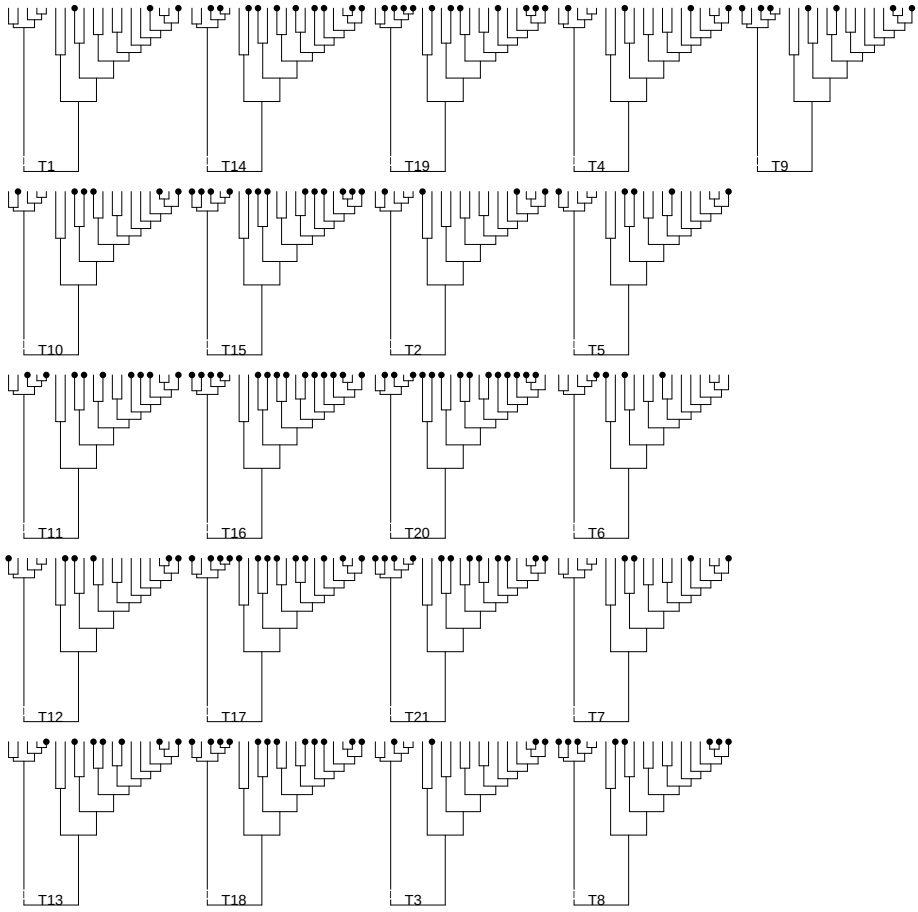
```
# ---Distribution of species assigned to treatments on regional phylogeny
spa = as.spacodi(t(treat))

## Formatting assumed to be that used with picante

phy.dotplot(spa, phy = amy.nature, tips.adj = c(0.5, 0.55), lab.adj = c(0, 1))

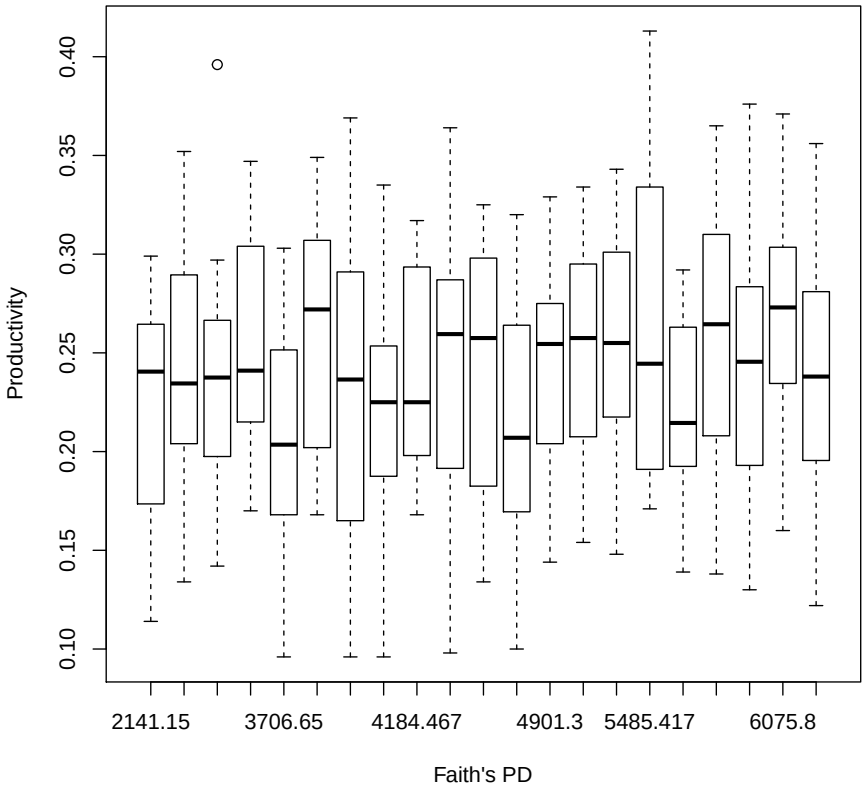
## Warning:  Tip labels are unlikely to plot properly with this many
trees.
```

Figure 2: Regional phylogeny showing the combination of species present in each of the 21 species compositions manipulated in the experiment

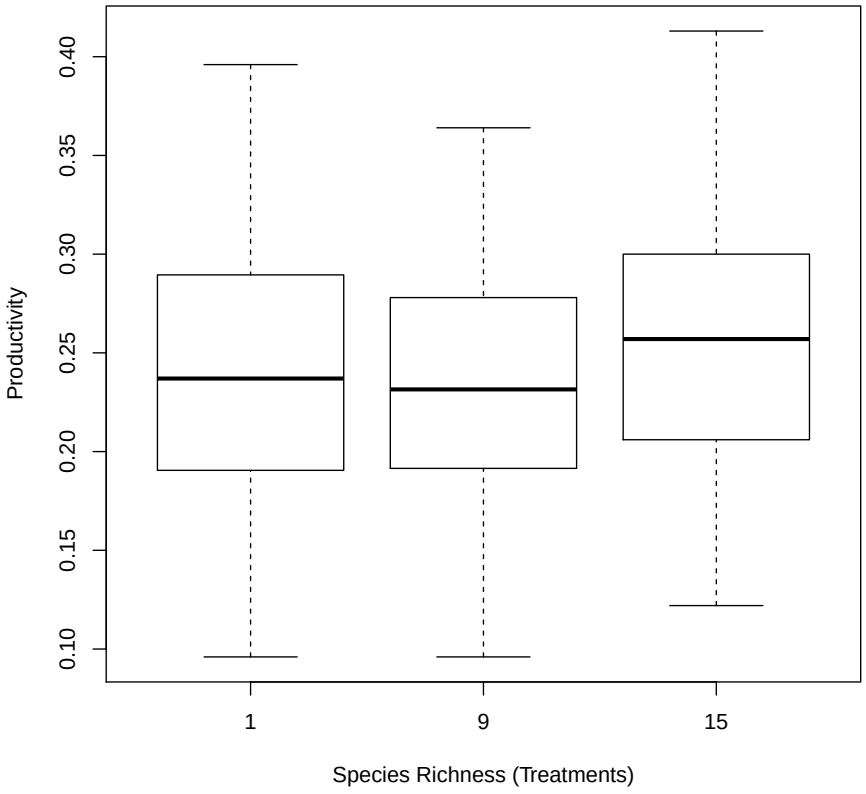


Exploratory data analysis

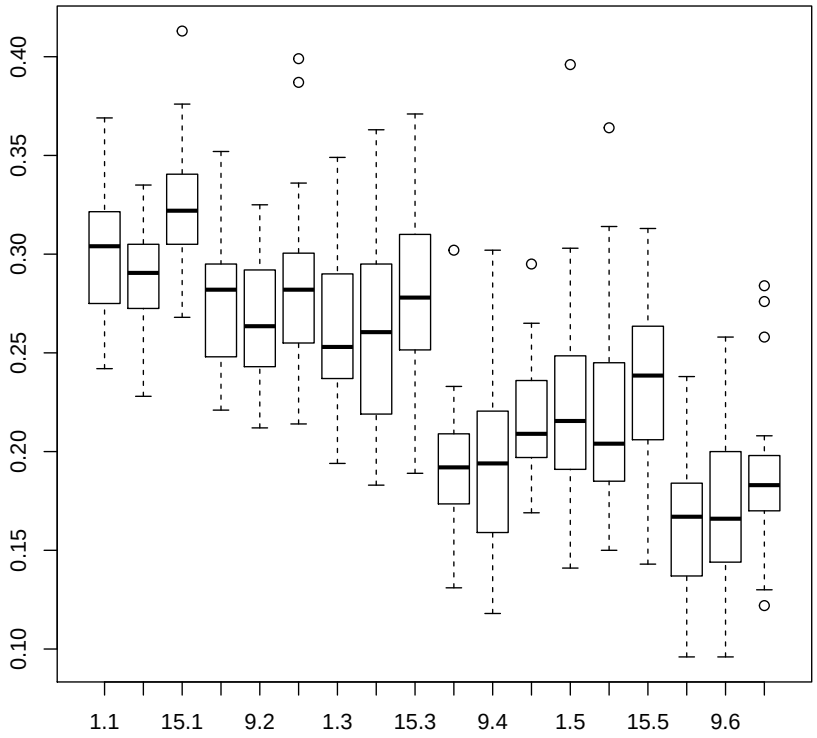
```
plot(prod ~ as.factor(PD), xlab = "Faith's PD", ylab = "Productivity")
```



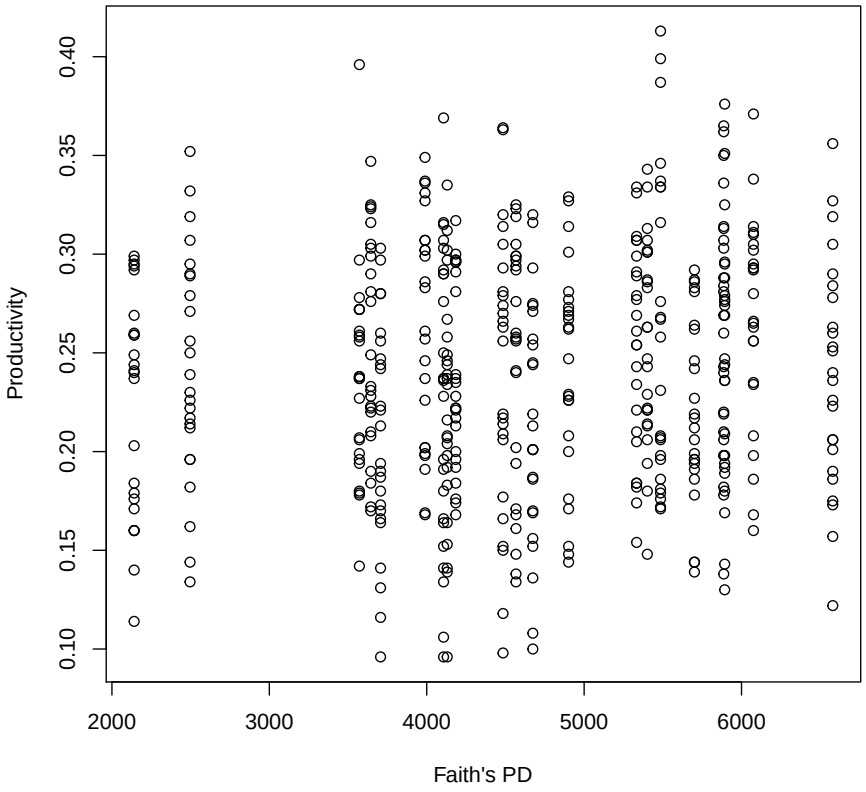

```
plot(prod ~ as.factor(div), xlab = "Species Richness (Treatments)", ylab = "Productivity")
```



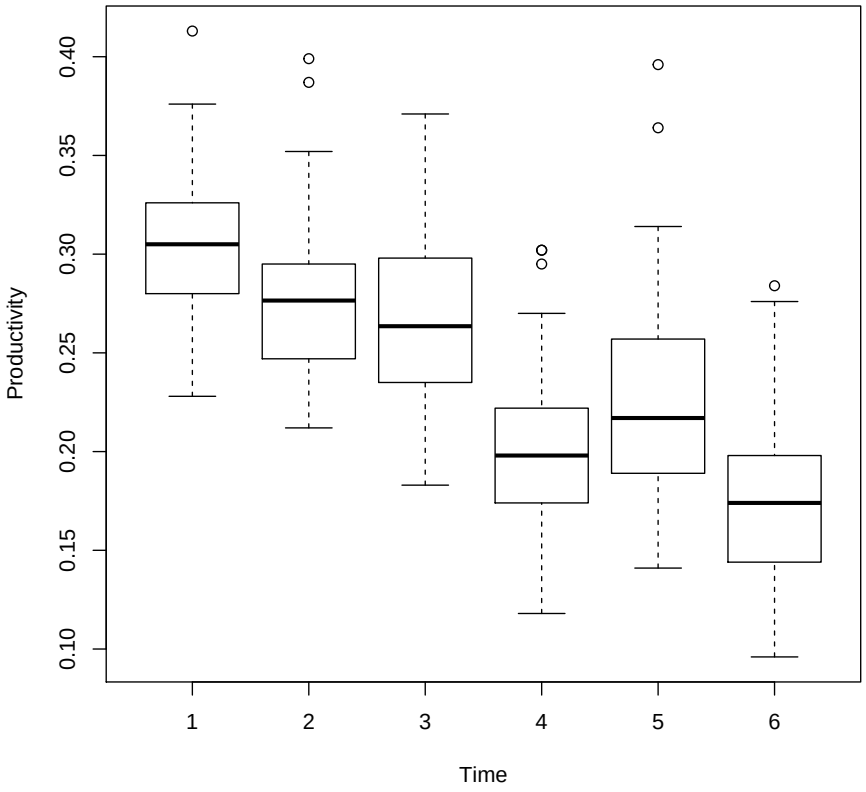

```
boxplot(prod ~ as.factor(div) + time)
```



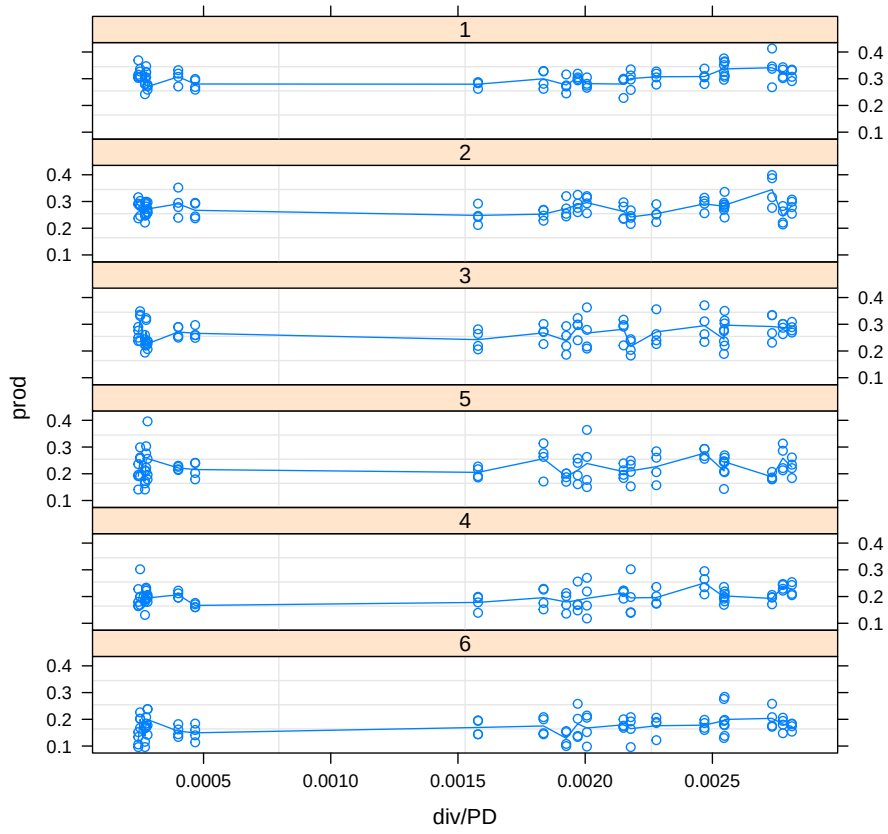

```
plot(prod ~ PD, xlab = "Faith's PD", ylab = "Productivity")
```




```
boxplot(prod ~ as.factor(time), xlab = "Time", ylab = "Productivity")
```




```
results <- groupedData(prod ~ div/PD | time, data = dados, FUN = mean)
plot(results)
```



Models for the influence of PD on ecosystem functioning

I run a rm-ANOVA for nested designs to investigate the influence of PD on productivity, decomposition, and respiration PD, nested within each treatment (=species richness).

```
## --- Linear Model with nested design and repeated measures for Productivity
mode <- lm(prod ~ trata + trata/COMP + TIME, data = dados)
mod1 <- anova(mode)
mode2 <- lm(prod ~ trata + PD + TIME, data = dados)
mod2 <- anova(mode2)
pf(mod1[1, 3]/mod1[3, 3], 2, 18, lower.tail = F)  #valor de P para efeito do tratamento (d
```

```
## [1] 0.0207
```

```
## -- Quadratic Model with nested design and repeated measures for
## Productivity
modeq <- lm(prod ~ trata + PD + I(PD^2) + TIME, data = dados)  #model with PD as a quadrat
anova(modeq)  #the quadratic term is significant
```

```
## Analysis of Variance Table
##
## Response: prod
##            Df Sum Sq Mean Sq F value  Pr(>F)
## trata       2  0.047  0.0235   15.65 2.6e-07 ***
## PD          1  0.000  0.0001    0.05   0.829
## I(PD^2)     1  0.007  0.0069    4.57   0.033 *
## TIME        5  1.052  0.2104  139.91 < 2e-16 ***
## Residuals 494  0.743  0.0015
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(modeq)  #to get the parameters and estimated values of the model

##
## Call:
## lm(formula = prod ~ trata + PD + I(PD^2) + TIME, data = dados)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.09443 -0.02632 -0.00164  0.02397  0.17350
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.57e-01   2.43e-02   10.60  < 2e-16 ***
## trata9      -5.84e-03   5.89e-03   -0.99   0.3226
## trata15      2.47e-02   9.15e-03    2.70   0.0072 **
## PD           2.28e-05   1.14e-05    2.00   0.0466 *
## I(PD^2)     -2.85e-09   1.34e-09   -2.14   0.0330 *
## TIME2       -2.92e-02   5.98e-03   -4.88  1.4e-06 ***
## TIME3       -3.67e-02   5.98e-03   -6.14  1.7e-09 ***
## TIME4       -1.04e-01   5.98e-03  -17.41  < 2e-16 ***
## TIME5       -7.98e-02   5.98e-03  -13.34  < 2e-16 ***
## TIME6       -1.32e-01   5.98e-03  -22.04  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0388 on 494 degrees of freedom
## Multiple R-squared:  0.598,Adjusted R-squared:  0.591
## F-statistic: 81.7 on 9 and 494 DF,  p-value: <2e-16

pred.mod <- predict(modeq)  #predicted values for each observation
dados$pred <- pred.mod  #binding to the data frame

me$pred <- aggregate(dados$pred, list(trata, PD), "mean")$x

mq <- lm(prod ~ PD + I(PD^2), data = dados)  #to get the values without time, to make the
# of the quadratic model
anova(mq)
```

```
## Analysis of Variance Table
##
## Response: prod
##            Df Sum Sq Mean Sq F value Pr(>F)
## PD          1  0.017 0.01705    4.66  0.031 *
## I(PD^2)     1  0.001 0.00054    0.15  0.701
## Residuals 501  1.832 0.00366
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(mq)
```

```
##
## Call:
## lm(formula = prod ~ PD + I(PD^2), data = dados)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.14141 -0.04707  0.00086  0.04576  0.16793
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.29e-01   3.39e-02    6.76    4e-11 ***
## PD          -8.39e-07   1.57e-05   -0.05     0.96
## I(PD^2)      6.75e-10   1.76e-09    0.38     0.70
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0605 on 501 degrees of freedom
## Multiple R-squared:  0.00951,Adjusted R-squared:  0.00556
## F-statistic: 2.41 on 2 and 501 DF,  p-value: 0.0912
```

```
coef(mq)
```

```
## (Intercept)          PD     I(PD^2)
##   2.294e-01  -8.392e-07   6.754e-10
```

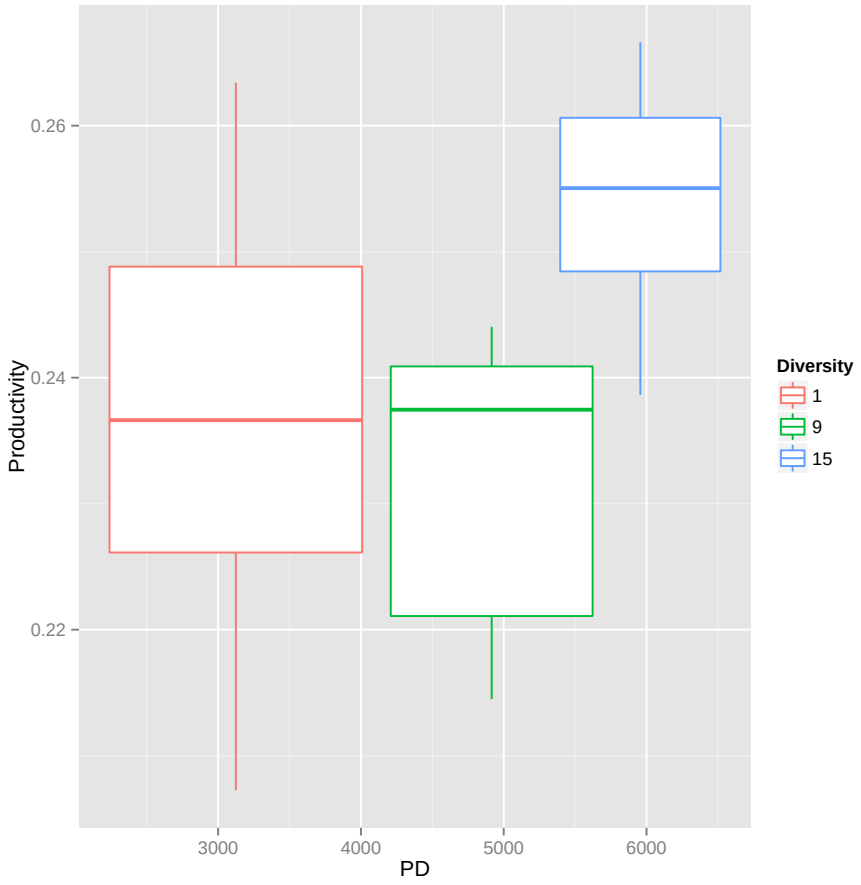
```
# to make the plot
names(me) = c("Diversity", "PD", "Productivity", "pred")  #renaming the columns to make th
me$pred <- coef(mq)[1] + coef(mq)[2] * me$PD + coef(mq)[3] * me$PD^2  #adding the predicte

# width represents PD
ggplot(me, aes(x = PD, y = Productivity, colour = Diversity)) + geom_line()
```

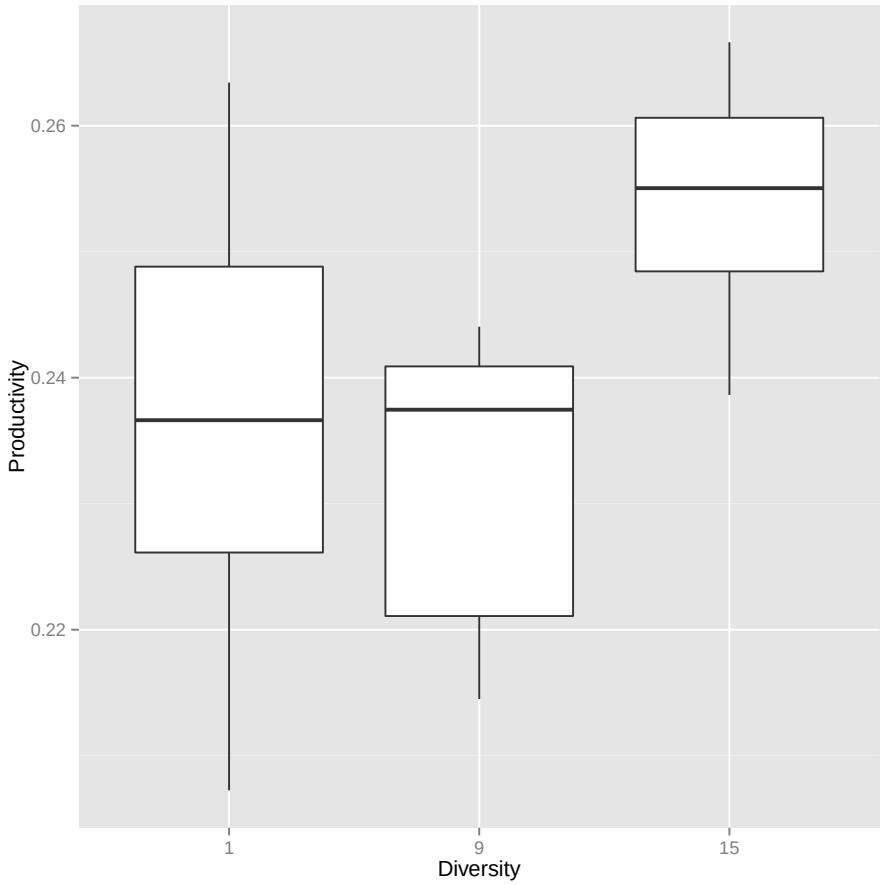



```
ggplot(me, aes(x = PD, y = Productivity, colour = Diversity)) + geom_boxplot()

## Warning: position_dodge requires constant width: output may be
incorrect
## Warning: position_dodge requires non-overlapping x intervals
```

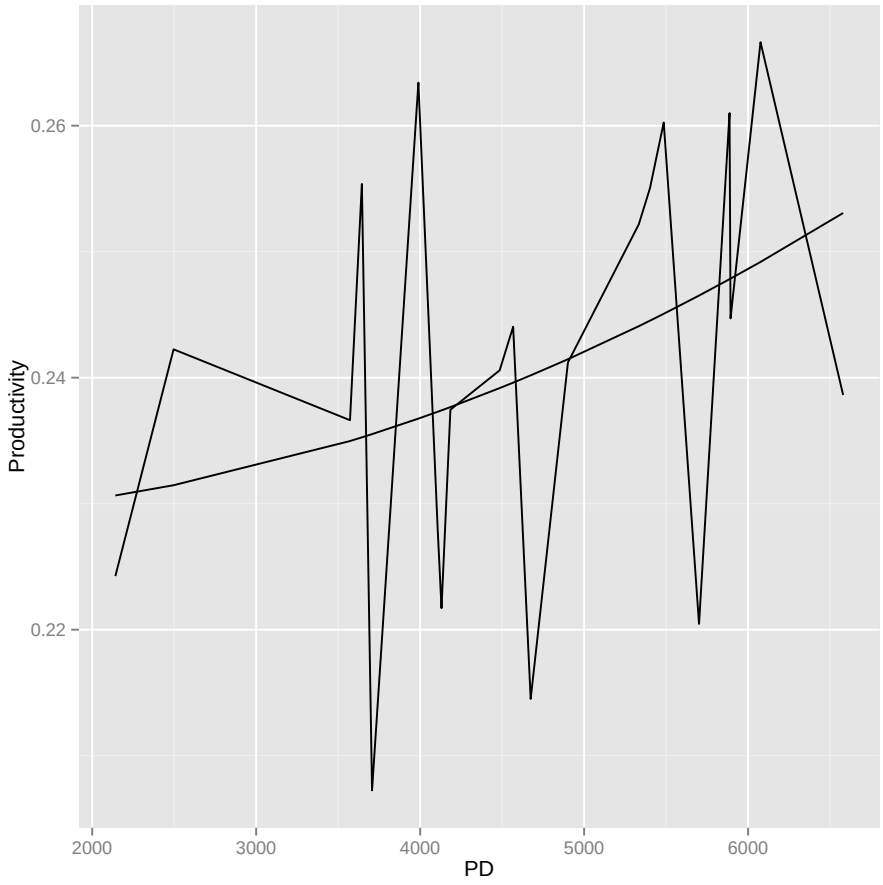



```
# same version but bw, instead
ggplot(me, aes(x = Diversity, y = Productivity)) + geom_boxplot()
```

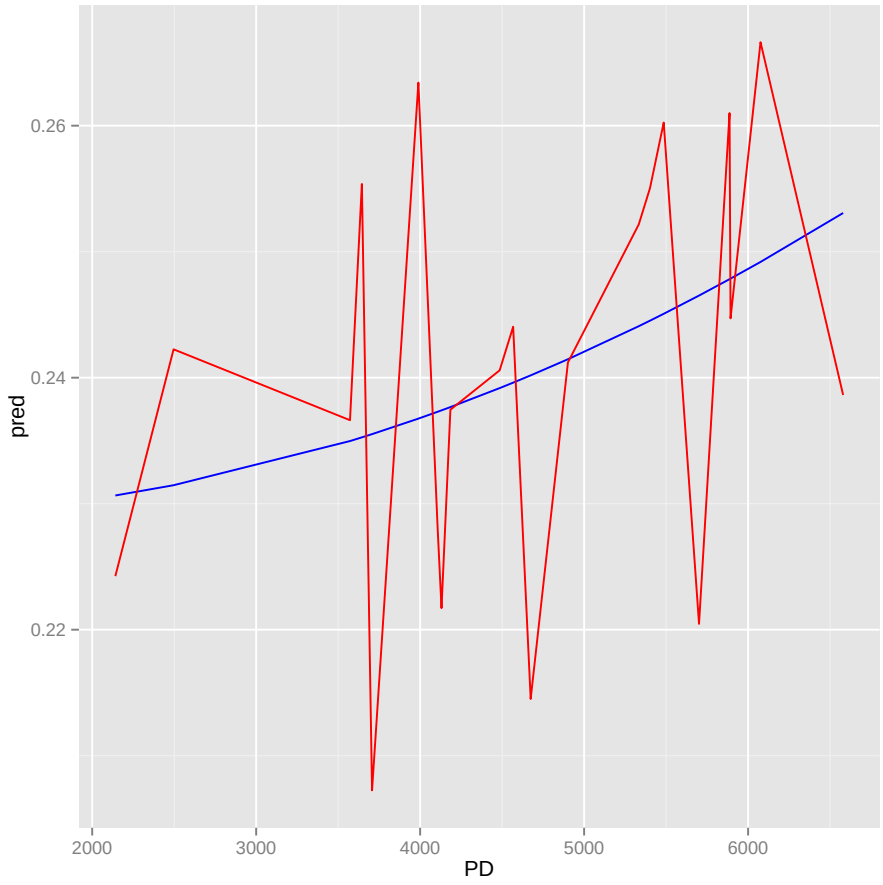



```
gf <- as.data.frame(cbind(rep(1:2, each = 21), rbind(cbind(me$PD, me$Productivity),
    cbind(me$PD, me$pred))))
names(gf) = c("Group", "PD", "Productivity")

ggplot(gf, aes(x = PD, y = Productivity, group = Group)) + geom_line()
```




```
# Final plot showing the variation in PD in red and the quadratic model in
# blue
ggplot(me, aes(x = PD)) + geom_line(aes(y = pred), colour = "blue") + geom_line(aes(y = Pr
    colour = "red")
```

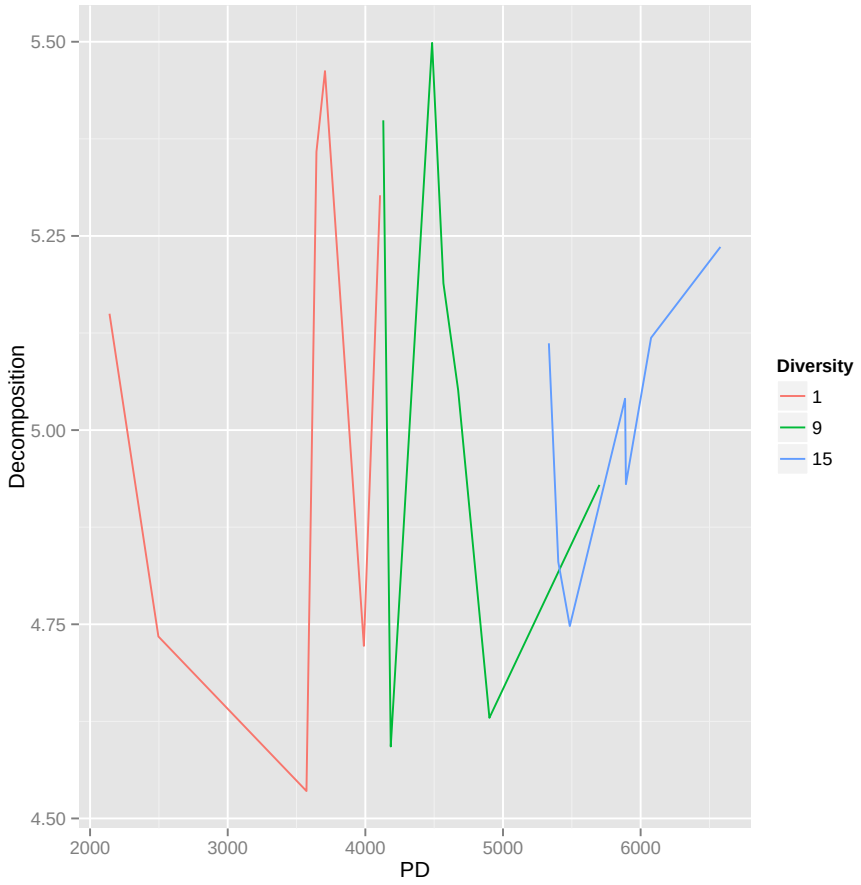



```
# same version but bw, instead
gf <- as.data.frame(cbind(rep(1:2, each = 21), rbind(cbind(me$PD, me$Productivity),
    cbind(me$PD, me$pred))))
names(gf) = c("Group", "PD", "Productivity")
```

```
modeL <- lm(dc ~ trata + trata/COMP + TIME, data = dados)
modee1 <- anova(modeL)
modee2 <- lm(dc ~ trata + PD + TIME, data = dados)
md2 <- anova(modee2)
pf(modee1[1, 3]/modee1[3, 3], 2, 18, lower.tail = F)  #valor de P para efeito do tratament

## [1] 0.9665
```

```
ggplot(dec.plot, aes(x = PD, y = Decomposition, colour = Diversity)) + geom_line()
```




```
m.resp.lin <- lm(resp ~ trata + PD + TIME, data = dados)
res.resp.lin <- anova(m.resp.lin)
pf(res.resp.lin[1, 3]/res.resp.lin[3, 3], 2, 18, lower.tail = F)  #valor de P para efeito

## [1] 0.9121

mdeq <- lm(resp ~ trata + PD + I(PD^2) + TIME, data = dados)  #model with PD as a quadrati
anova(mdeq)  #the quadratic term is significant

## Analysis of Variance Table
##
## Response: resp
##             Df Sum Sq Mean Sq F value  Pr(>F)
## trata        2  0.023  0.0114   13.38 2.2e-06 ***
## PD           1  0.001  0.0013    1.58   0.209
## I(PD^2)      1  0.005  0.0047    5.53   0.019 *
## TIME         5  0.614  0.1228  144.73 < 2e-16 ***
## Residuals 494  0.419  0.0008
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
modq <- lm(resp ~ PD + I(PD^2), data = dados)  #to get the values without time, to make th
anova(modq)
```

```
## Analysis of Variance Table
##
## Response: resp
##            Df Sum Sq Mean Sq F value Pr(>F)
## PD          1  0.014 0.01351    6.46  0.011 *
## I(PD^2)     1  0.000 0.00002    0.01  0.920
## Residuals 501  1.048 0.00209
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(modq)
```

```
##
## Call:
## lm(formula = resp ~ PD + I(PD^2), data = dados)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.10238 -0.03473 -0.00556  0.03457  0.20891
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.62e-01   2.57e-02    6.30  6.4e-10 ***
## PD           5.74e-06   1.19e-05    0.48     0.63
## I(PD^2)     -1.33e-10   1.33e-09   -0.10     0.92
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0457 on 501 degrees of freedom
## Multiple R-squared:  0.0127,Adjusted R-squared:  0.0088
## F-statistic: 3.23 on 2 and 501 DF,  p-value: 0.0403
```

```
coef(modq)
```

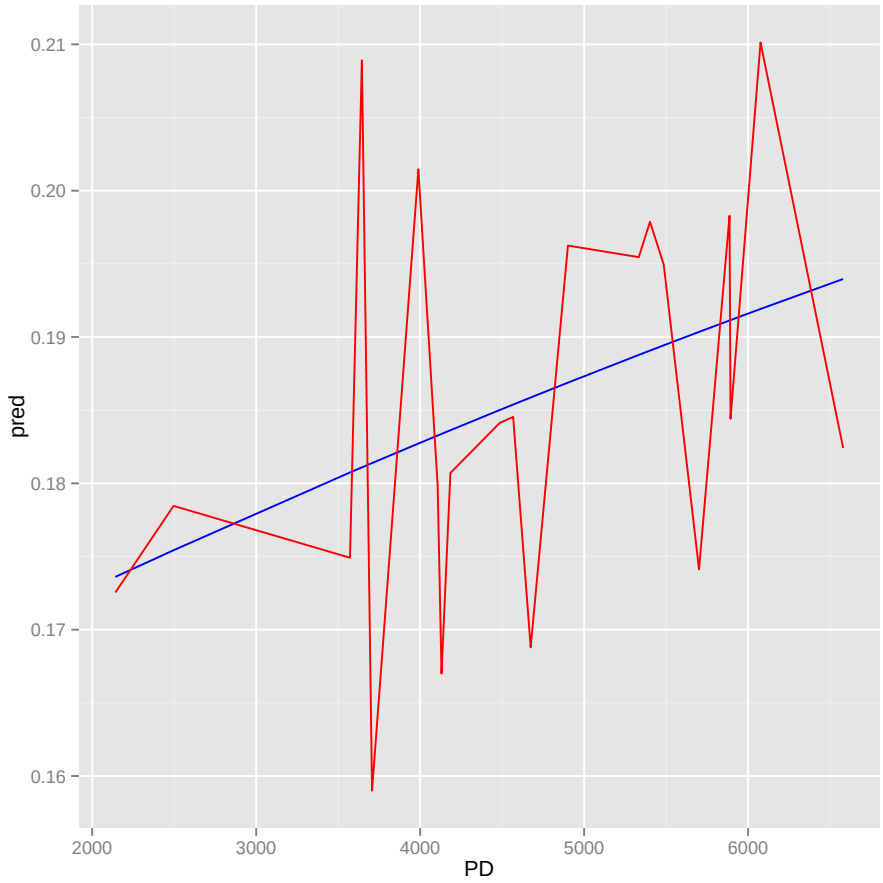
```
## (Intercept)          PD     I(PD^2)
##   1.619e-01   5.743e-06  -1.327e-10
```

```
resp.pred <- predict(modq)  #predicted values for each observation
dados$resp.pred <- resp.pred  #binding to the data frame

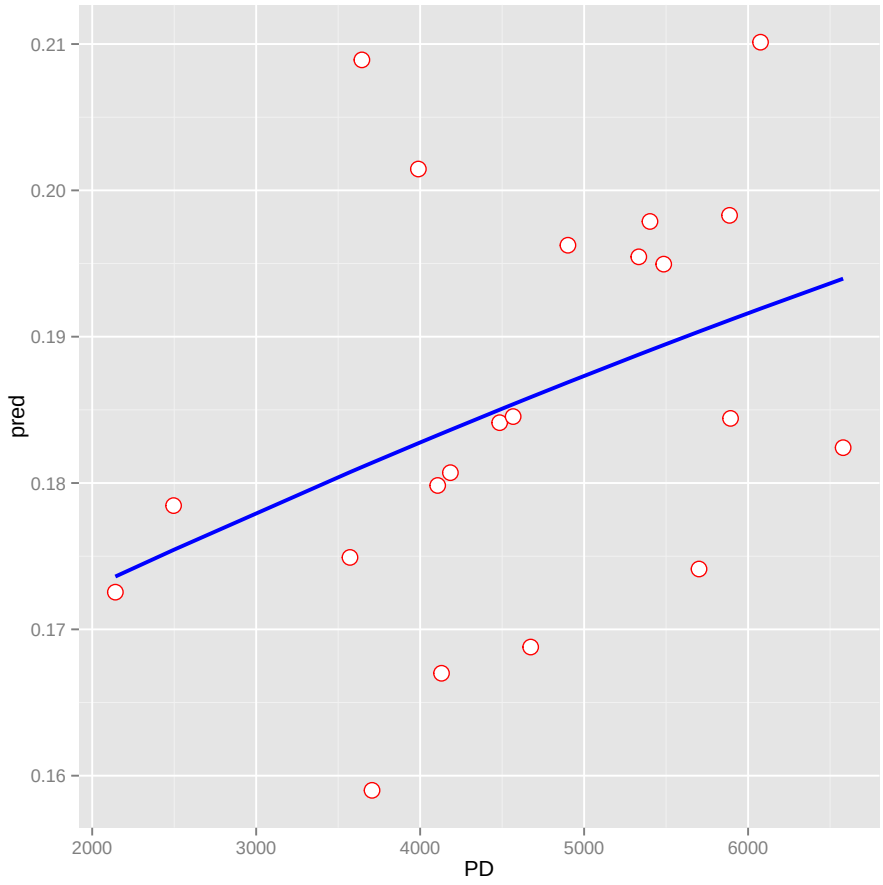
resp.plot$resp.pred <- aggregate(dados$resp.pred, list(trata, PD), "mean")$x

resp.plot$resp.pred <- coef(modq)[1] + coef(modq)[2] * resp.plot$PD + coef(modq)[3] *
    resp.plot$PD^2  #adding the predicted values
# for PD by the quadratic model
names(resp.plot) = c("Diversity", "PD", "Respiration", "pred")  #renaming the columns to m
```

```
# Final plot showing the variation in PD in red and the quadratic model in
# blue
ggplot(resp.plot, aes(x = PD)) + geom_line(aes(y = pred), colour = "blue") +
    geom_line(aes(y = Respiration), colour = "red")
```




```
# Just another version with data as points and model as line
ggplot(resp.plot, aes(x = PD)) + geom_line(aes(y = pred), colour = "blue", linetype = "sol
    size = 1) + geom_point(aes(y = Respiration), colour = "red", size = 4, shape = 21,
    fill = "white")
```



Model Selection for PD variables

Now, I entered all alternative diversity variables in a model selection to see which
one best describes the three EF.

Productivity

```
# -with PD
prod.PD <- lm(prod ~ trata + PD + TIME, data = dados)
mq.prodPD <- lm(prod ~ trata + PD + I(PD^2) + TIME, data = dados)

# -With MPD
prod.pd <- lm(prod ~ trata + mpd + TIME, data = d)
mq.mpd <- lm(prod ~ trata + mpd + I(mpd^2) + TIME, data = d)

# -With Haed
prod.haed <- lm(prod ~ trata + haed + TIME, data = d)
mq.haed <- lm(prod ~ trata + haed + I(haed^2) + TIME, data = d)
```

```r
# -With Rao
prodrao <- lm(prod ~ trata + raodiv + TIME, data = d)
mq.rao <- lm(prod ~ trata + raodiv + I(raodiv^2) + TIME, data = d)

# -With MNTD
prod.mntd <- lm(prod ~ trata + mntd + TIME, data = d)
mq.mntd <- lm(prod ~ trata + mntd + I(mntd^2) + TIME, data = d)

# -With Simpson Phylogenetic
prod.SimpPhyl <- lm(prod ~ trata + SimpsonPhyl + TIME, data = d)
mq.SimpPhyl <- lm(prod ~ trata + SimpsonPhyl + I(SimpsonPhyl^2) + TIME, data = d)

# -With Shannon
prod.Shannon <- lm(prod ~ trata + Shannon + TIME, data = d)
mq.Shannon <- lm(prod ~ trata + Shannon + I(Shannon^2) + TIME, data = d)

# -With Simpson
prod.Simpson <- lm(prod ~ trata + Simpson + TIME, data = d)
mq.Simpson <- lm(prod ~ trata + Simpson + I(Simpson^2) + TIME, data = d)

# -With Simpson
prod.Intens_Q <- lm(prod ~ trata + Intens_Q + TIME, data = d)
mq.Intens_Q <- lm(prod ~ trata + Intens_Q + I(Intens_Q^2) + TIME, data = d)

# -- Model selection table
(ICtab(prod.PD, mq.prodPD, prod.pd, mq.mpd, prod.haed, mq.haed, prodrao, mq.rao,
    prod.mntd, mq.mntd, prod.SimpPhyl, mq.SimpPhyl, prod.Shannon, mq.Shannon,
    prod.Simpson, mq.Simpson, prod.Intens_Q, mq.Intens_Q, type = c("AICc"),
    weights = TRUE, delta = TRUE, sort = TRUE, nobs = 504))
```

```
##               dAICc df weight
## mq.prodPD       0.0 11 0.78
## prod.PD         2.6 10 0.22
## prod.mntd     434.7 10 <0.001
## prod.Simpson  434.9 10 <0.001
## prod.Intens_Q 434.9 10 <0.001
## prod.pd       435.1 10 <0.001
## prod.Shannon  435.1 10 <0.001
## prod.haed     435.2 10 <0.001
## mq.mpd        435.3 11 <0.001
## prodrao       436.1 10 <0.001
## prod.SimpPhyl 436.1 10 <0.001
## mq.Intens_Q   436.3 11 <0.001
## mq.haed       436.7 11 <0.001
## mq.mntd       436.7 11 <0.001
## mq.Shannon    436.8 11 <0.001
## mq.Simpson    437.0 11 <0.001
## mq.rao        438.2 11 <0.001
## mq.SimpPhyl   438.2 11 <0.001
```

Respiration

```
# -with PD
resp.PD <- lm(resp ~ trata + PD + TIME, data = dados)
qpd <- lm(resp ~ trata + PD + I(PD^2) + TIME, data = dados)

# -With MPD
resp.mpd <- lm(resp ~ trata + mpd + TIME, data = d)
qmpd <- lm(resp ~ trata + mpd + I(mpd^2) + TIME, data = d)

# -With Haed
resp.haed <- lm(resp ~ trata + haed + TIME, data = d)
qhaed <- lm(resp ~ trata + haed + I(haed^2) + TIME, data = d)

# -With Rao
resprao <- lm(resp ~ trata + raodiv + TIME, data = d)
qrao <- lm(resp ~ trata + raodiv + I(raodiv^2) + TIME, data = d)

# -With MNTD
resp.mntd <- lm(resp ~ trata + mntd + TIME, data = d)
qmntd <- lm(resp ~ trata + mntd + I(mntd^2) + TIME, data = d)

# -With Simpson Phylogenetic
resp.SimpPhyl <- lm(resp ~ trata + SimpsonPhyl + TIME, data = d)
qSimpPhyl <- lm(resp ~ trata + SimpsonPhyl + I(SimpsonPhyl^2) + TIME, data = d)

# -With Shannon
resp.Shannon <- lm(resp ~ trata + Shannon + TIME, data = d)
qShannon <- lm(resp ~ trata + Shannon + I(Shannon^2) + TIME, data = d)

# -With Simpson
resp.Simpson <- lm(resp ~ trata + Simpson + TIME, data = d)
qSimpson <- lm(resp ~ trata + Simpson + I(Simpson^2) + TIME, data = d)

# -With Simpson
resp.Intens_Q <- lm(resp ~ trata + Intens_Q + TIME, data = d)
qIntens_Q <- lm(resp ~ trata + Intens_Q + I(Intens_Q^2) + TIME, data = d)

# -- Model selection table
(ICtab(resp.PD, qpd, qmntd, resp.mntd, qrao, resprao, qhaed, resp.haed, qmpd,
    resp.mpd, resp.SimpPhyl, qSimpPhyl, resp.Shannon, qShannon, resp.Intens_Q,
    qIntens_Q, type = c("AICc"), weights = TRUE, delta = TRUE, sort = TRUE,
    nobs = 504))

##               dAICc df weight
## qpd             0.0 11 0.85
## resp.PD         3.5 10 0.15
## resp.mpd      447.8 10 <0.001
## resprao       448.5 10 <0.001
```

```
## resp.SimpPhyl 448.5 10 <0.001
## qmpd          449.9 11 <0.001
## resp.haed     450.1 10 <0.001
## resp.Shannon  450.1 10 <0.001
## resp.mntd     450.1 10 <0.001
## resp.Intens_Q 450.1 10 <0.001
## qrao          450.4 11 <0.001
## qSimpPhyl     450.4 11 <0.001
## qIntens_Q     451.4 11 <0.001
## qmntd         451.6 11 <0.001
## qhaed         451.9 11 <0.001
## qShannon      452.1 11 <0.001
```

Decomposition

```
# -with PD
dec.PD <- lm(dc ~ trata + PD + TIME, data = dados)
qdec <- lm(dc ~ trata + PD + I(PD^2) + TIME, data = dados)

# -With MPD
dec.mpd <- lm(dc ~ trata + mpd + TIME, data = d)
decqmpd <- lm(dc ~ trata + mpd + I(mpd^2) + TIME, data = d)

# -With Haed
dc.haed <- lm(dc ~ trata + haed + TIME, data = d)
qdhaed <- lm(dc ~ trata + haed + I(haed^2) + TIME, data = d)

# -With Rao
dcrao <- lm(dc ~ trata + raodiv + TIME, data = d)
qdrao <- lm(dc ~ trata + raodiv + I(raodiv^2) + TIME, data = d)

# -With MNTD
dc.mntd <- lm(dc ~ trata + mntd + TIME, data = d)
qdmntd <- lm(dc ~ trata + mntd + I(mntd^2) + TIME, data = d)

# -With Simpson Phylogenetic
dc.SimpPhyl <- lm(dc ~ trata + SimpsonPhyl + TIME, data = d)
qdSimpPhyl <- lm(dc ~ trata + SimpsonPhyl + I(SimpsonPhyl^2) + TIME, data = d)

# -With Shannon
dc.Shannon <- lm(dc ~ trata + Shannon + TIME, data = d)
qdShannon <- lm(dc ~ trata + Shannon + I(Shannon^2) + TIME, data = d)

# -With Simpson
dc.Simpson <- lm(dc ~ trata + Simpson + TIME, data = d)
qdSimpson <- lm(dc ~ trata + Simpson + I(Simpson^2) + TIME, data = d)

# -With Simpson
```

```r
dc.Intens_Q <- lm(dc ~ trata + Intens_Q + TIME, data = d)
qdIntens_Q <- lm(dc ~ trata + Intens_Q + I(Intens_Q^2) + TIME, data = d)

# -- Model selection table
(ICtab(dec.PD, qdec, qdmntd, dc.mntd, qdrao, dcrao, qdhaed, dc.haed, decqmpd,
    dec.mpd, dc.SimpPhyl, qdSimpPhyl, dc.Shannon, qdShannon, dc.Intens_Q, qIntens_Q,
    type = c("AICc"), weights = TRUE, delta = TRUE, sort = TRUE, nobs = 504))
```

```
##             dAICc df weight
## qIntens_Q      0  11 1
## dec.PD      2910  10 <0.001
## qdec        2912  11 <0.001
## dc.mntd     3079  10 <0.001
## qdmntd      3081  11 <0.001
## dc.haed     3082  10 <0.001
## dc.Shannon  3082  10 <0.001
## dec.mpd     3082  10 <0.001
## dc.Intens_Q 3082  10 <0.001
## qdhaed      3084  11 <0.001
## qdrao       3084  11 <0.001
## qdSimpPhyl  3084  11 <0.001
## qdShannon   3084  11 <0.001
## dcrao       3084  10 <0.001
## dc.SimpPhyl 3084  10 <0.001
## decqmpd     3084  11 <0.001
```

Plotting the best model

```r
qdIntens_Q <- lm(dc ~ Intens_Q + I(Intens_Q^2), data = d)
anova(qdIntens_Q)
```

```
## Analysis of Variance Table
##
## Response: dc
##                Df Sum Sq Mean Sq F value Pr(>F)
## Intens_Q        1      2   2.285    2.43   0.12
## I(Intens_Q^2)   1      1   0.833    0.89   0.35
## Residuals     501    471   0.939
```

```r
summary(qdIntens_Q)
```

```
##
## Call:
## lm(formula = dc ~ Intens_Q + I(Intens_Q^2), data = d)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -5.440 -0.556 -0.054  0.525  4.529
##
## Coefficients:
```

```
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.90e+00   8.32e-02   58.93   <2e-16 ***
## Intens_Q       1.14e-05   8.19e-06    1.39     0.16
## I(Intens_Q^2) -1.35e-10   1.43e-10   -0.94     0.35
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.969 on 501 degrees of freedom
## Multiple R-squared:  0.00658,Adjusted R-squared:  0.00262
## F-statistic: 1.66 on 2 and 501 DF,  p-value: 0.191

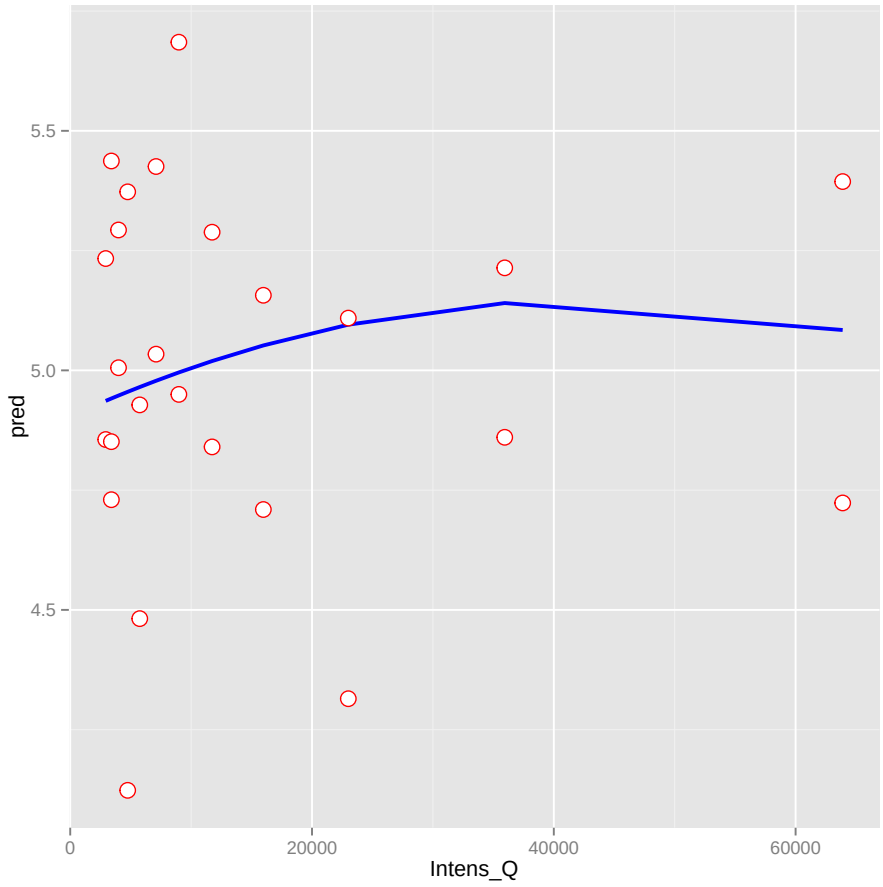
coef(qdIntens_Q)

##   (Intercept)      Intens_Q I(Intens_Q^2)
##     4.904e+00     1.142e-05    -1.345e-10

dec.pred <- predict(qdIntens_Q)  #predicted values for each observation
d$dec.pred <- dec.pred  #binding to the data frame

dec.plot1 <- aggregate(dc, list(trata, d$Intens_Q), "mean")
dec.plot1$dec.pred <- aggregate(d$dec.pred, list(trata, d$Intens_Q), "mean")$x
names(dec.plot1) = c("Diversity", "Intens_Q", "Decompositon", "pred")  #renaming the colum
dec.plot1$dec.pred <- coef(qdIntens_Q)[1] + coef(qdIntens_Q)[2] * dec.plot1$Intens_Q +
    coef(qdIntens_Q)[3] * dec.plot1$Intens_Q^2  #adding the predicted values
# for PD by the quadratic model

# Just another version with data as points and model as line
ggplot(dec.plot1, aes(x = Intens_Q)) + geom_line(aes(y = pred), colour = "blue",
    linetype = "solid", size = 1) + geom_point(aes(y = Decompositon), colour = "red",
    size = 4, shape = 21, fill = "white")
```


5.5
5.0
4.5
pred
0
20000
40000
60000
Intens_Q